

Performance comparison of sampling techniques with machine learning algorithms for churn prediction in telecommunication

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ABSTRACT

The telecommunications industry faces intense competition in customer retention, making churn prediction a key area of research in machine learning. A systematic churn prediction model is essential, as large datasets, high-dimensional features, and class imbalance can hinder prediction performance. This study proposes a novel churn prediction model using machine learning techniques, evaluated on a dataset of 7043 telecom customers. It explores various sampling techniques such as the Synthetic Minority Oversampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN), and Conditional Tabular Generative Adversarial Network (CTGAN) combined with machine learning models to address class imbalance in churn prediction. Experimental results demonstrate that CTGAN, when paired with a Weighted Random Forest (WRF) classifier, consistently outperforms other methods, achieving a remarkable accuracy of 99.79%, along with strong performance in terms of precision, recall, F1-score, and AUC. These findings highlight CTGAN's effectiveness in addressing dataset imbalance and enhancing the generalizability of churn prediction models, offering a valuable solution for the telecommunications industry.

1. Introduction

Businesses thrive in competitive markets by prioritizing customer satisfaction [1,2]. In such dynamic environments with numerous alternatives, dissatisfied customers are quick to switch to competitors [3]. This phenomenon, known as customer churn, can be triggered by various factors, including product or service issues, price increases, poor quality, limited features, or other concerns [4]. Industries such as banking, airlines, and telecommunications are particularly vulnerable to the negative impacts of high customer churn [5]. As a result, companies especially those in the telecom industry are striving to build and maintain long-lasting relationships with their customers.

The telecom industry's rapid expansion and diversity have made the market highly competitive [6]. Intense competition, market saturation, rapid technological advancements, and attractive offers from rivals contribute to significant customer churn in the industry. In this fast-paced and demanding environment, maximizing profits is crucial [7,8]. Strategies such as increasing customer lifetime value, fostering customer loyalty, and acquiring new clients have been proposed to achieve this goal. Several studies have shown that retaining existing customers is generally more cost-effective than acquiring new ones [9,

10]. Accurately predicting customer attrition is therefore essential for effective customer retention campaigns [11,12].

Customer Churn Prediction (CCP) is essential for developing retention strategies that increase revenue and strengthen industry reputation. In the telecommunications sector and other business-driven domains, CCP research is particularly valuable, as it contributes to revenue growth, fosters innovation, improves service quality, enhances customer satisfaction, optimizes resource allocation, and supports technological adaptation [8,9]. To ensure sustainable growth and long-term profitability, telecommunications companies must prioritize this area due to the industry's dynamic and unpredictable nature. Today, telecom companies have access to vast amounts of customer data, including call records, text messages, voicemails, and customer profiles. These data are invaluable for identifying customers at risk of churning, making accurate forecasting of customer behavior critical [13,14]. There are two primary approaches to reducing customer attrition: reactive and proactive strategies. A reactive strategy involves recognizing a customer's intention to leave and then offering incentives to retain them [7,12]. In contrast, a proactive strategy anticipates potential churn and takes early action by offering personalized incentives to promote retention. Proactive churn management is commonly framed as a binary classification

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task aimed at distinguishing churners from non-churners [14,15].

A number of approaches, including machine learning (ML) methods and rule-based systems, have been explored to address CCP [16]. However, rule-based CCP techniques are often considered inflexible and difficult to scale [6,17]. In contrast, ML-based approaches have been widely adopted, demonstrating varying levels of performance. The primary challenges stem from the unique characteristics of churn datasets, which can significantly impact the effectiveness of conventional ML algorithms [18–20]. The quality and structure of the input data play a critical role in determining the performance of ML models. Therefore, it is essential to ensure that datasets used for CCP are well-structured and properly prepared for analysis. One of the most pressing challenges is class imbalance, which can greatly affect the accuracy of ML models. This issue arises when the dataset contains an unequal number of instances in each class typically with far more non-churners than churners. As a result, the minority class (churners) is often underrepresented, leading to biased models that favor the majority class while underperforming on the minority class. Addressing class imbalance is crucial for building effective ML-based CCP models [15,21]. It occurs when there is a substantial disparity between the number of samples in the majority and minority classes. Unevenly distributed class labels can result in errors and significant challenges when training ML models [22–24].

Krasic et al. [25] addressed fraud detection using real telecom data with high class imbalance by applying SMOTE and evaluating various machine learning (ML) classifiers, which led to improved model performance. Alif et al. [26] enhanced churn prediction using the Telco Churn dataset by applying both oversampling and undersampling techniques with Random Forest, Naive Bayes, and SVM, resulting in notable gains in specificity (up to 65.41 % for SVM). Ly et al. [27] developed a kernel SVM-based churn prediction model, incorporating feature selection methods (Sequential Forward Selection and Sequential Backward Selection) along with resampling techniques (SMOTE-Tomek and SMOTE-ENN), achieving an F1-score of 99 % and an accuracy of 98.9 %.

Ileberi et al. [28] tackled credit card fraud detection by applying ADASYN for class balancing and Recursive Feature Elimination with Cross-Validation (RFECV) for feature selection. Their approach achieved high Matthews Correlation Coefficient (MCC) scores using XGBoost (0.9994) and Random Forest (0.9991), while LightGBM showed a 194 % improvement in MCC. Halim et al. [29] used the Ecoli dataset to evaluate the performance of Random Forest with SMOTE and ADASYN, achieving an 84 % F1-score and 90 % balanced accuracy, and identified balanced accuracy as a more suitable metric for imbalanced datasets. Peng et al. [30] proposed a GA-XGBoost model for telecom churn prediction, integrating ADASYN and Genetic Algorithm (GA) for optimization, and incorporating SHAP for model interpretability. The model outperformed traditional methods and provided strong interpretability through SHAP values.

Sana et al. [31] developed a privacy-preserving churn prediction framework using Generative Adversarial Networks (GANs) and adaptive Weight-of-Evidence encoding, achieving up to a 28.9 % improvement in accuracy and a 27.9 % gain in F-measure. Chang et al. [32] employed ensemble models—most notably Random Forests, which achieved 91.66 % accuracy and leveraged explainable AI (XAI) tools such as LIME and SHAP to improve interpretability and support strategic retention decisions. Lee et al. [33] evaluated CTGAN and TGAN for EEG data augmentation in brain-computer interface (BCI) applications, finding that CTGAN produced data more similar to real EEG signals, although it did not significantly improve ML performance highlighting the need for more effective augmentation techniques. Han et al. [34] proposed a CTGAN-based framework combining tabular modeling with Gaussian mixture models to balance transient stability samples and enhance power system assessment on the IEEE 39-bus system.

This study investigates several sampling techniques to address class imbalance in a telecommunications dataset, including SMOTE,

ADASYN, and CTGAN. CTGAN is recognized as an advanced technique for creating synthetic data that effectively balances class distribution and enhances predictive performance. After testing various machine learning approaches, the Random Forest (RF) model demonstrated exceptional predictive performance and stability in noisy environments. LightGBM (LGBM) excels in both efficiency and predictive performance, making it highly effective for large datasets. XGBoost (XGB) achieves competitive results by efficiently capturing complex patterns. The Decision Tree (DT) provides a simple baseline with moderate performance. The Hybrid Stacking-based Logistic Regression (HSLR) method optimally integrates multiple algorithms to enhance prediction reliability. Hybrid Statistical YLinear Regression with RF (HSLR-RF) effectively handles nonlinear relationships while maintaining interpretability. The performance of the minority class was significantly improved using Weighted Random Forest (WRF), which assigns tailored weights to different classes. Our findings indicate that the combination of CTGAN and WRF achieves the best overall performance. However, the RF, LGBM, XGB, DT, HSLR, and HSLR-RF methods also yielded promising results, highlighting the importance of addressing class imbalance and exploring hybrid approaches for optimal outcomes.

2. Related works

Burez et al. [35] investigated under-sampling and advanced sampling techniques across datasets from the telecommunications, banking, and online retail sectors. Their findings highlighted that XGB and WRF outperformed standard Random Forests in handling imbalanced data. The research emphasized the crucial role of variable selection in enhancing the efficiency of the advanced CUBE sampling technique. Pun et al. [36] explored customer churn prediction (CP) in the telecom, banking, and insurance industries, employing ADASYN to mitigate class imbalance. They achieved an AUC of 0.90 and an F1-score of 0.68 with their ensemble model, which combined SVM, LR, and backpropagation.

Maw et al. [37] also addressed class imbalance in telecom churn prediction by utilizing SMOTE and ADASYN while evaluating algorithms such as RF, LightGBM, and XGBoost. They recommended that future research be extended to larger datasets and incorporate fairness metrics. Imani et al. [38] applied SMOTE to churn prediction in the telecom sector, assessing advanced models including CatBoost, LightGBM, and XGBoost. Their results demonstrated CatBoost's superior performance. The study suggests further exploration of techniques like ADASYN on highly imbalanced datasets to enhance prediction accuracy.

Adiputra et al. [39] proposed CTGAN-ENN, which combines CTGAN with ENN to address imbalance and overlap in churn data, improving model performance and reducing runtime across five datasets. Kate et al. [40] introduced ChaoticGAN, leveraging chaotic noise for oversampling in churn, fraud, and loan prediction. It outperformed existing GANs in AUC and, when combined with OCSVM undersampling, showed further gains. Adiputra et al. [41] developed CostLearnGAN, a cost-sensitive GAN that enhances machine learning model performance on imbalanced churn datasets, achieving the highest mean rank (1.44) and robustness score (5.68) across six datasets. Raghu et al. [42] utilized CTGAN to generate realistic network traffic in SDN-based cybersecurity, achieving 97 % accuracy with a decision tree and demonstrating its value for IoT and smart city applications.

Watanuki et al. tackled challenges in survey research such as high costs, rare population sampling, and privacy concerns by augmenting small datasets using DGNNs (CTGAN, TVAE, CopulaGAN). Testing 11 datasets, they found that augmentation boosted predictive performance, with CopulaGAN best preserving variable dependencies. However, discrepancies in feature importance between real and synthetic data require further refinement [43]. Fang et al. predicted employee turnover using an ML pipeline (SMOTE, GAN, RFE) on HR data. The Random Forest model with GAN-balanced data achieved the best performance (accuracy: 0.98, F1-score: 0.95). Validation on IBM HR data confirmed generalizability, with XGBoost and RF+DNN also performing well [44].

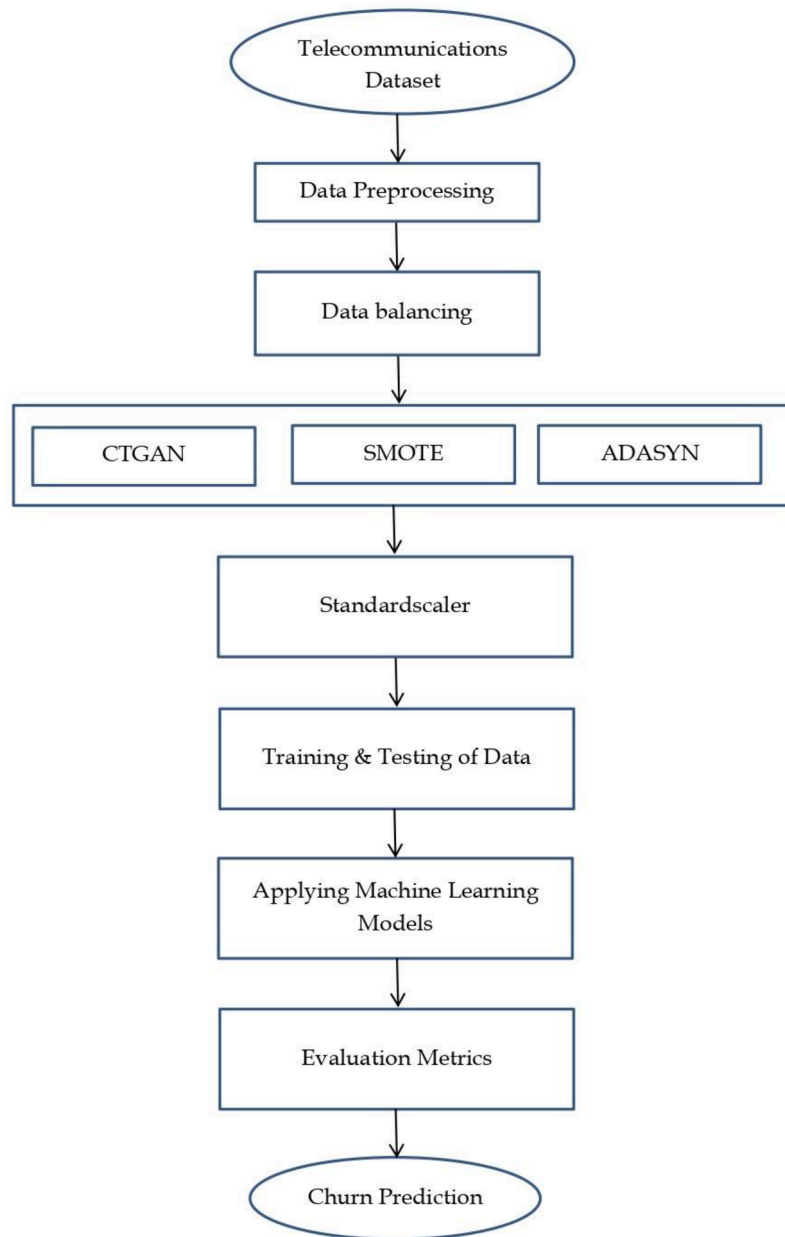


Fig. 1. Churn prediction framework for telecommunications dataset.

Khatiman et al. [45] examined synthetic data generation for 5 G networks using test drive data, leveraging CTGAN and TVAE to meet AI/ML data requirements. While TVAE achieved the highest overall performance (94.14 %), CTGAN better preserved certain data distributions. In their study, sampling techniques were applied to train classifiers, and various machine learning algorithms were evaluated using different sets of important features.

CTGAN is a type of GAN specifically designed to generate synthetic tabular data, particularly when the data includes both numerical and categorical features. Unlike traditional GANs, CTGAN effectively addresses imbalanced datasets by employing a conditional generation mechanism that ensures underrepresented categories are learned and synthesized more accurately. It does this by conditionally sampling training data based on specific values of categorical columns, enabling the generator to focus more on minority classes and produce realistic, balanced synthetic data. The results indicated that this approach outperformed several well-known classifiers.

3. Dataset characteristics

The dataset used in this study is a publicly available telecommunications customer churn dataset from Kaggle (<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>), containing 7043 samples. The target variable, Churn, is binary, with 1869 churned customers (approximately 26.5 %) and 5174 non-churned customers (approximately 73.5 %), indicating a class imbalance. The dataset includes 21 features, comprising both numerical variables (e.g., tenure, Monthly Charges, Total Charges) and categorical variables (e.g., gender, Internet Service, Contract, Payment Method). Categorical attributes describe customer demographics, service usage, and account information, such as Senior Citizen, Partner, Online Security, and Streaming TV, among others. This mix of structured data provides a rich foundation for training machine learning models for churn prediction.

4. Research methodology

The dataset used in this study contains a binary target variable (Churn), which was initially imbalanced, with churned customers representing a small minority. It includes both categorical features (e.g., Contract, Payment Method) and numerical features (e.g., Monthly Charges, tenure), which were preprocessed using label encoding for categorical variables and standardized using StandardScaler for numerical ones. All models were trained using a standard 80:20 train-test split to ensure consistent evaluation. Fig. 1 illustrates the preprocessing pipeline and sampling strategy used to address class imbalance and prepare the data for model training. SMOTE was selected for its efficiency, broad library support, and strong performance, using `sampling_strategy='auto'`, `random_state=42`, and `k_neighbors=5`. ADASYN, applied with the same parameters, yielded notable metric gains and proved especially effective in refining decision boundaries for more complex classifiers. CTGAN, implemented via CTGANSynthesizer, was tuned with 300 epochs, a batch size of 500, a learning rate of $2e-4$, and hidden layers of (128, 128). It produced statistically similar synthetic data, resulting in a balanced dataset that improved model performance and validated CTGAN's effectiveness in handling class imbalance in churn prediction tasks. The RF model was configured with 100 trees and no depth limit; to address class imbalance, appropriate settings were applied, and a `random_state` of 42 was used across all models to ensure reproducibility. XGB was set with 100 estimators, a learning rate of 0.1, a depth of 6, and `scale_pos_weight=1`. DT were limited to `max_depth=6` with `criterion='gini'`, while SVC employed an RBF kernel with `probability=True`. LR, also using `random_state=42`, served as the final estimator in a Stacking Classifier that combined SVC and DT, with `passthrough=True` to include the original features. This hybrid ensemble improved generalization, reduced overfitting, and enhanced classification performance across models.

5. Sampling techniques

CTGAN, SMOTE, and ADASYN were selected for their proven effectiveness in addressing class imbalance in tabular datasets. CTGAN is particularly well-suited for modeling complex data distributions using deep generative techniques. SMOTE [46] generates synthetic samples through interpolation, helping to reduce overfitting, while ADASYN concentrates on harder-to-learn regions to refine decision boundaries. Both SMOTE and ADASYN are widely adopted oversampling techniques that have demonstrated strong empirical performance in prior churn prediction studies. These methods outperform basic resampling strategies by generating more informative samples and preserving the underlying data structure, ultimately improving model generalization on imbalanced datasets.

5.1. Synthetic minority over-sampling technique

SMOTE is a widely used technique for addressing class imbalance by generating synthetic minority class samples through linear interpolation between existing instances and their k -nearest neighbors in the feature space [47]. The following equation illustrates this interpolation:

$$x_{\text{new}} = x_i + \lambda \times (x_{\text{nn}} - x_i). (1)$$

Where x_i = minority class instance; λ = random number; $x_{\text{nn}} = k$ -nearest neighbors; SMOTE effectively expands the minority class in the feature space while preserving the data distribution, reducing the risk of overfitting associated with simple replication. Its efficiency, broad library support, and strong performance made it well-suited for churn prediction. Using parameters such as `sampling_strategy='auto'`, `random_state=42`, and `k_neighbors=5`, SMOTE consistently improved both the F1-score and ROC AUC across models, confirming that balanced data had a greater impact on performance than model complexity. Thus,

SMOTE proved to be an optimal method for handling class imbalance in churn prediction.

5.2. Adaptive synthetic sampling approach

ADASYN, introduced by He et al. [48], is an advanced oversampling technique that adaptively generates synthetic samples near decision boundaries where minority instances are sparse. Unlike SMOTE, it adjusts the number of synthetic samples per instance based on the level of learning difficulty, focusing more on harder-to-learn cases. This enhances data balance, reduces model bias, and improves classification performance across models such as neural networks, decision trees, and SVMs [48]. Its ability to refine decision boundaries makes ADASYN a valuable tool for imbalanced predictive modeling. The following approach generates a synthetic sample x_{syn} for a given minority instance x_i :

$$x_{\text{syn}} = x_i + \lambda \times (x_{\text{NN}} - x_i). (2)$$

Where x_i = minority class instance; λ = random number; $x_{\text{NN}} = k$ -nearest neighbors. ADASYN improves classifier sensitivity to minority class patterns and reduces bias toward the majority class. Implemented with parameters similar to those used in SMOTE (`sampling_strategy='auto'`, `random_state=42`, `n_neighbors=5`), it enhanced models such as Random Forest and Decision Trees, boosting the F1-score by 18–22 % and the AUC by 12–15 %. Its ability to refine decision boundaries makes it particularly effective for complex models like SVMs and neural networks, confirming ADASYN as a strong choice for handling class imbalance in predictive modeling.

5.3. Conditional generative adversarial network

CTGAN is a GAN-based technique specifically designed for generating synthetic data to rectify class imbalance in datasets. Using an encoder-decoder architecture, the generator creates synthetic samples by mapping the source data into a latent space [49]. To reduce the loss function, an algorithm is trained to generate synthetic samples that resemble the original data. The discriminator is trained to separate the real data from the synthetic data by optimizing the loss function.

$$LD = \{-\log(D(X)) - \log(1 - D(G(Z)))\}. (3)$$

CTGAN is a type of Generative Adversarial Network (GAN) specifically tailored for generating synthetic data to tackle class imbalance issues in datasets. It utilizes an encoder-decoder framework in which the generator transforms input data into a latent space to produce synthetic samples [48]. The CTGAN-based synthetic data generation involves initializing the model with 300 epochs and a batch size of 500. The generator learns to mimic the real data distribution by minimizing its loss, while the discriminator distinguishes between real and synthetic samples. In this study, CTGAN was implemented using CTGANSynthesizer, fine-tuned with a learning rate of $2e-4$ and two hidden layers of size (128, 128), ensuring stable convergence. The model is trained on real tabular data, capturing complex feature relationships, especially in imbalanced datasets. Once trained, it generates synthetic samples that reflect the statistical properties of the original data. Post-processing includes validating distributional similarity, removing outliers, and reapplying necessary transformations for downstream use.

6. Classification of algorithms

This section explores the application of various machine learning models for churn prediction, including RF, LGBM, XGB, HSLR, HSLR-RF, and WRF. All models are trained using a standard 80:20 train-test split to ensure consistent evaluation. Before training, the features are normalized using StandardScaler to bring them to a common scale, which improves model performance and convergence particularly for algorithms

sensitive to feature magnitudes, such as logistic regression. Each classifier is evaluated to assess its effectiveness in accurately predicting customer churn.

6.1. Random forest algorithm

RF is an ensemble learning technique that constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. It operates by sampling several subsets of the training data with replacement—a process known as bootstrap aggregating (bagging). Each decision tree is trained on a different subset of the data, and at every split, only a randomly selected subset of features is considered rather than all available features. For classification tasks, the final prediction is determined by majority voting, while for regression tasks, it is based on the average prediction. The prediction of a Random Forest model for classification is defined by the following equation:

$$\hat{y} = \underset{k}{\operatorname{argmax}} \sum_{m=1}^M 1(h_m(x) = k) \quad (4)$$

where $h_m(x)$ represents the prediction from the m^{th} DT, k is the output label, M represents the total count of trees and $1(\cdot)$ is an indicator function that counts votes for each class. The regression prediction is computed as the mean of the outputs from all trees.

$$\hat{y} = \frac{1}{M} \sum_{m=1}^M h_m(x) \quad (5)$$

Where $h_m(x)$ denotes the predicted value from the m^{th} tree. We trained the Random Forest model with 100 trees and no depth limit, applied techniques to address class imbalance, and set `random_state=42` for reproducibility. These settings ensure consistency, enhance generalizability, and provide a solid baseline for evaluating sampling techniques.

6.2. Light gradient boosting machine algorithm

The LGBM framework is designed for maximum efficiency, incorporating techniques such as leaf-wise tree growth, histogram-based feature binning, and Gradient-based One-Side Sampling (GOSS) to ensure scalability and fast execution. Unlike the traditional level-wise tree growth used in Gradient Boosting Decision Trees (GBDT), LightGBM splits the leaf with the highest loss reduction, resulting in deeper and more optimized trees. The model aims to minimize the following objective function:

$$L^{(t)} \approx \sum_{i=1}^n \left[l(y_i, F_{t-1}(x_i)) + g f_t(x_i) + \frac{1}{2} h f_t^2(x_i) \right] + \Omega(f_t) \quad (6)$$

where g_i and h_i are the gradient and Hessian of the loss function. The model updates iteratively as:

$$F_t(x) = F_{t-1}(x) + f_t(x) \quad (7)$$

and selects splits based on maximum loss reduction:

$$\Delta L = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (8)$$

Here, G and H represent the sum of gradients and Hessians for the child nodes. These optimizations make LightGBM significantly faster and more memory-efficient than traditional GBDT, making it particularly well-suited for large-scale machine learning applications. The model was configured with 100 estimators, a learning rate of 0.1, unlimited depth, and `class_weight='balanced'` to address class imbalance. This configuration yielded stable and reliable performance during evaluation.

6.3. Extreme gradient boosting algorithm

XGB is a scalable and highly effective gradient boosting technique that enhances traditional GBDT through the incorporation of regularization, shrinkage, and parallel processing, resulting in superior performance. It builds decision trees sequentially, with each tree correcting the errors of its predecessor by minimizing the following objective function:

$$L^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{j=1}^T \Omega(f_j) \quad (9)$$

In this case $l(y_i, \hat{y}_i)$ represent the loss function, $\Omega(f_j) = \gamma T + \frac{1}{2} \lambda \sum w_j^2$. Eqs. (7) and (8) reflect similar gradient boosting principles in XGB and LGBM; however, LGBM differs through its use of histogram-based feature binning, leaf-wise tree growth, and Gradient-based One-Side Sampling (GOSS). XGB was configured with 100 estimators, a learning rate of 0.1, a maximum depth of 6, the logloss evaluation metric, and `scale_pos_weight=1` to address class imbalance, enabling stable and interpretable model evaluation.

6.4. Decision tree algorithm

A supervised learning technique called a DT recursively splits the dataset based on feature values for tasks such as regression and classification. It has a tree-like structure, where each internal node represents a decision rule, each branch corresponds to an outcome, and each leaf node represents a class label (for classification) or a numerical value (for regression). To determine the best split at each node, the algorithm uses criteria such as Mean Squared Error (MSE) for regression, and Entropy (Information Gain) or Gini impurity for classification. The Gini impurity is defined as follows:

$$\text{Gini} = 1 - \sum_{i=1}^c p_i^2 \quad (10)$$

Here, p_i denotes the probability associated with class i . The Entropy is calculated as:

$$\text{Entropy} = - \sum_{i=1}^c p_i \log_2 p_i \quad (11)$$

Where p_i signifies the relative frequency of class i in a node. The measure of entropy reduction, known as information gain, is given by:

$$\text{IG} = \text{Entropy}(\text{parent}) - \sum_j \frac{|S_j|}{|S|} \text{Entropy}(s_j) \quad (12)$$

where S_j are the subsets after a split. We used a DT classifier with criterion='gini', `max_depth=6`, and `random_state=42` to balance simplicity with control over overfitting. This configuration ensures reproducibility and provides interpretability of the model's decision-making process, with potential for future extension to ensemble methods.

6.5. Hybrid stacking-based logistic regression algorithm

The HSLR is an ensemble method that combines several base models through stacking, with LR serving as the meta-model for final predictions. Stacking involves training diverse base learners (e.g., DT, SVM, RF, XGB) and using their predictions as new input features for the meta-learner. LR is defined as follows:

$$P(Y = 1|X) = \frac{1}{1 + e^{-\left(\beta_0 + \sum_{i=1}^n \beta_i X_i\right)}} \quad (13)$$

Table 1

Table of descriptive data analysis.

	Count	Mean	Std	Min	25 %	50 %	75 %	Max
gender	7043.0	0.4952	0.5000	0.00	0.0	0.00	1.00	1.00
SeniorCitizen	7043.0	0.1621	0.3686	0.00	0.0	0.00	0.00	1.00
Partner	7043.0	0.4830	0.4997	0.00	0.0	0.00	1.00	1.00
Dependents	7043.0	0.2996	0.4581	0.00	0.0	0.00	1.00	1.00
tenure	7043.0	32.3711	24.5595	0.00	9.0	29.00	55.00	72.00
PhoneService	7043.0	0.9032	0.2958	0.00	1.0	1.00	1.00	1.00
MultipleLines	7043.0	0.6155	0.6560	0.00	0.0	1.00	1.00	2.00
InternetService	7043.0	1.2229	0.7789	0.00	1.0	1.00	2.00	2.00
OnlineSecurity	7043.0	0.7200	0.7969	0.00	0.0	1.00	1.00	2.00
OnlineBackup	7043.0	0.7782	0.7785	0.00	0.0	1.00	1.00	2.00
DeviceProtection	7043.0	0.7772	0.7788	0.00	0.0	1.00	1.00	2.00
TechSupport	7043.0	0.7236	0.7959	0.00	0.0	1.00	1.00	2.00
StreamingTV	7043.0	0.8177	0.7632	0.00	0.0	1.00	1.00	2.00
StreamingMovies	7043.0	0.8212	0.7617	0.00	0.0	1.00	1.00	2.00
Contract	7043.0	0.6905	0.8338	0.00	0.0	1.00	1.00	2.00
PaperlessBilling	7043.0	0.5922	0.4915	0.00	0.0	1.00	1.00	2.00
PaymentMethod	7043.0	1.3156	1.1489	0.00	0.0	1.00	2.00	3.00
MonthlyCharges	7043.0	64.7617	30.0900	18.25	35.5	70.35	89.85	118.75
Churn	7043.0	0.2654	0.4416	0.00	0.0	0.00	1.00	1.00

Where B_0 is the intercept and B_i are the model coefficients, acts as the final classifier by learning an optimal combination of base model outputs. The hybrid stacking model combined SVC (RBF kernel, probability=True), DT (max_depth=5), and LR (final estimator), all with random_state=42. Using passthrough=True, the model incorporated original features alongside base outputs, boosting robustness, reducing overfitting, and improving classification accuracy through ensemble synergy.

6.6. Hybrid statistical linear regression with the RF algorithm

A HSLR-RF algorithm combines the strengths of LR and RF to improve predictive accuracy and robustness in regression tasks. LR models relationships using the equation:

$$Y = \beta_0 + \sum_{i=1}^n \beta_i X_i + \epsilon \quad (14)$$

Here, Y stands for the dependent variable, X_i refers to the independent variables, β_i are the associated coefficients, and ϵ represents the error term. However, LR assumes a linear relationship, which limits its effectiveness in capturing complex patterns. To overcome this, RF regression, an ensemble of DT, is used to model nonlinear dependencies, predicting the target as:

$$\hat{Y} = \frac{1}{T} \sum_{t=1}^T f_t(X) \quad (15)$$

Where T represents the total number of trees, and $f_t(X)$ indicates the prediction from each tree. In this hybrid model, RF captures complex patterns, while LR refines the residuals, improving accuracy and reducing overfitting. Configured with n_estimators=100 and random_state=42, the approach ensures reproducibility, balanced learning, and effective churn prediction.

6.7. Weighted random forest algorithm

The WRF algorithm is an extension of the standard Random Forest that assigns weights to training samples, enabling the model to handle class imbalance and place greater emphasis on specific instances during the learning process. During the construction of each decision tree, a sample's weight influences its likelihood of being selected for splitting, ensuring that the model pays more attention to harder-to-predict or underrepresented classes. The algorithm minimizes the weighted error for each tree, and the final prediction is obtained by aggregating the

weighted outputs of the individual trees. The weighted error for a tree t can be calculated as:

$$\text{weightedError}_t = \frac{\sum_{i=1}^n w_i \cdot I(y_i \neq \hat{y}_i)}{\sum_{i=1}^n w_i} \quad (16)$$

Where w_i represents the weight of the i^{th} sample, y_i is the actual label, and $\hat{y}_i$ denotes the predicted label. The final prediction of the ensemble is obtained through a weighted majority vote for classification tasks or a weighted average for regression tasks. A Weighted Random Forest, configured with class_weight='balanced', n_estimators=100, and random_state=42, is used to improve minority class prediction in an imbalanced churn dataset. After encoding and scaling the features, the model is trained using stratified splitting and evaluated using standard metrics to ensure balanced and accurate performance.

7. Performance metrics

Performance assessment of the model was conducted through metrics like the F1-score, AUC, accuracy, recall, and precision, which are determined by the following parameters:

- True positive (TP): Capture instances where churning customers (positive) are classified correctly as churn (positive).
- True negative (TN): Indicates cases where nonchurning customers (negative) are classified correctly as not churning (negative).
- False positive (FP): Identifies cases where nonchurning customers (negative) are mistakenly classified as churn (positive).
- False negative (FN): Reflects cases where churning customers (positive) are wrongly classified as not churning (negative).

Accuracy: Accuracy quantifies the percentage of customers correctly classified as either churning or not churning. This is expressed in the form of:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

Precision: It evaluates the proportion of true positives among all cases that were predicted as positive, including both accurate and inaccurate predictions. It is calculated as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Table 2
Performance of algorithms without sampling method.

Classifier	Accuracy	Precision	Recall	F1	AUC
RF	79.28 %	65.28 %	46.38 %	54.23 %	68.75 %
LGBM	80.13 %	65.66 %	52.28 %	58.21 %	71.22 %
XGB	77.79 %	59.20 %	51.74 %	55.22 %	69.45 %
DT	72.11 %	47.50 %	50.94 %	49.16 %	65.33 %
HSLR	80.91 %	68.71 %	51.21 %	58.68 %	71.40 %
HSLR-RF	79.84 %	66.06 %	49.06 %	56.31 %	69.99 %
WRF	79.63 %	65.05 %	50.27 %	56.71 %	82.00 %

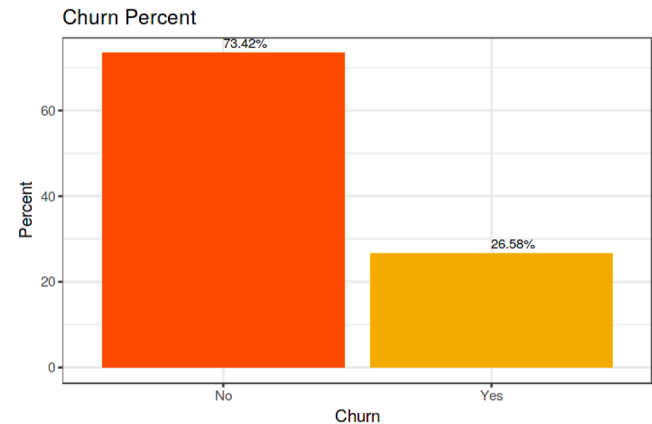


Fig. 2. Bar chart showing the distribution of customer churn.

Recall: The percentage of true positive cases that were accurately identified out of all actual positive instances is known as the recall. The formula for the recall is as follows:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 score: The F1 score combines precision and recall as the harmonic mean, with a maximum value of 1 representing optimal performance in both metrics. The mathematical representation is as follows:

$$\text{F1 Score} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

AUC: The AUC represents the area beneath the ROC curve, which is plotted with the True Positive Rate (TPR) on the vertical axis and the False Positive Rate (FPR) on the horizontal axis. TPR is calculated as TP / (TP + TN), while FPR is given by FP / (FP + TN). The AUC value serves as a reliable metric for assessing the overall performance of a bank’s customer churn prediction model. A higher AUC value, closer to 1, indicates better predictive performance of the model.

8. Results and discussion

8.1. Descriptive data analysis

To improve the clarity and interpretability of the dataset attributes, summary statistics were computed for all variables. The results are presented in Table 1.

The summary statistics in Table 2 provide an overview of the dataset’s features, comprising 7043 customer records. Most categorical variables, such as gender, SeniorCitizen, Partner, and Dependents, are binary, with mean values indicating class distributions (e.g., approximately 49.5 % of customers are female, and only 16.2 % are senior

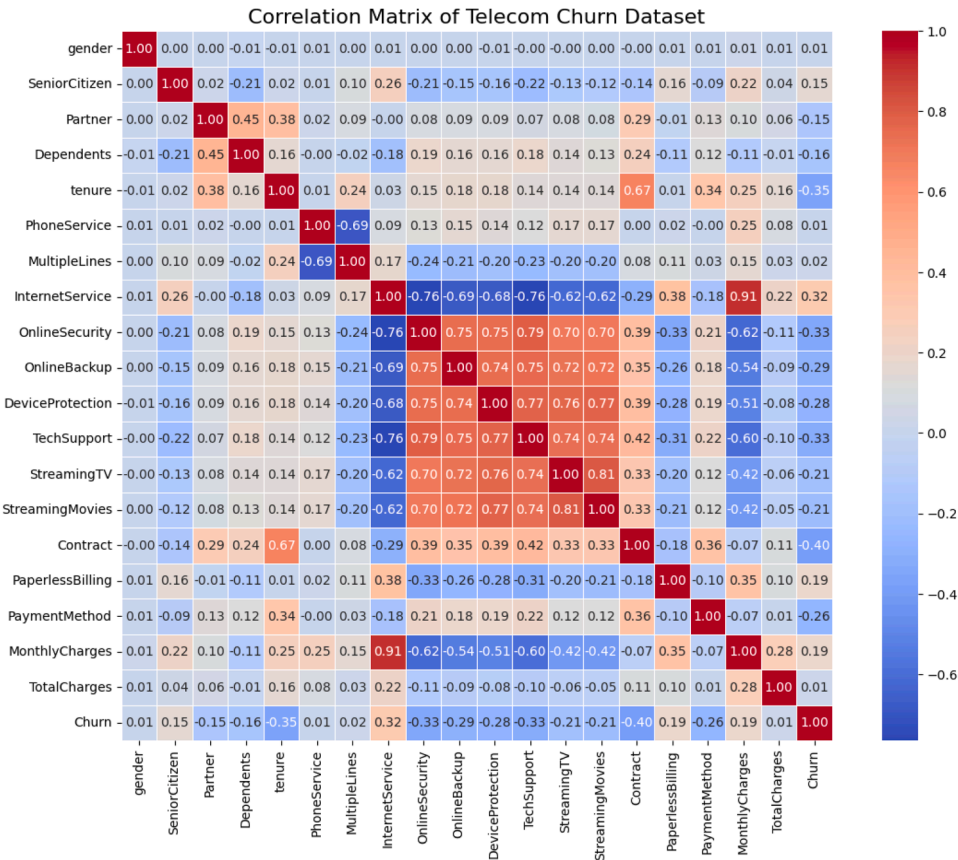


Fig. 3. Graphical representation of the Correlation Matrix.

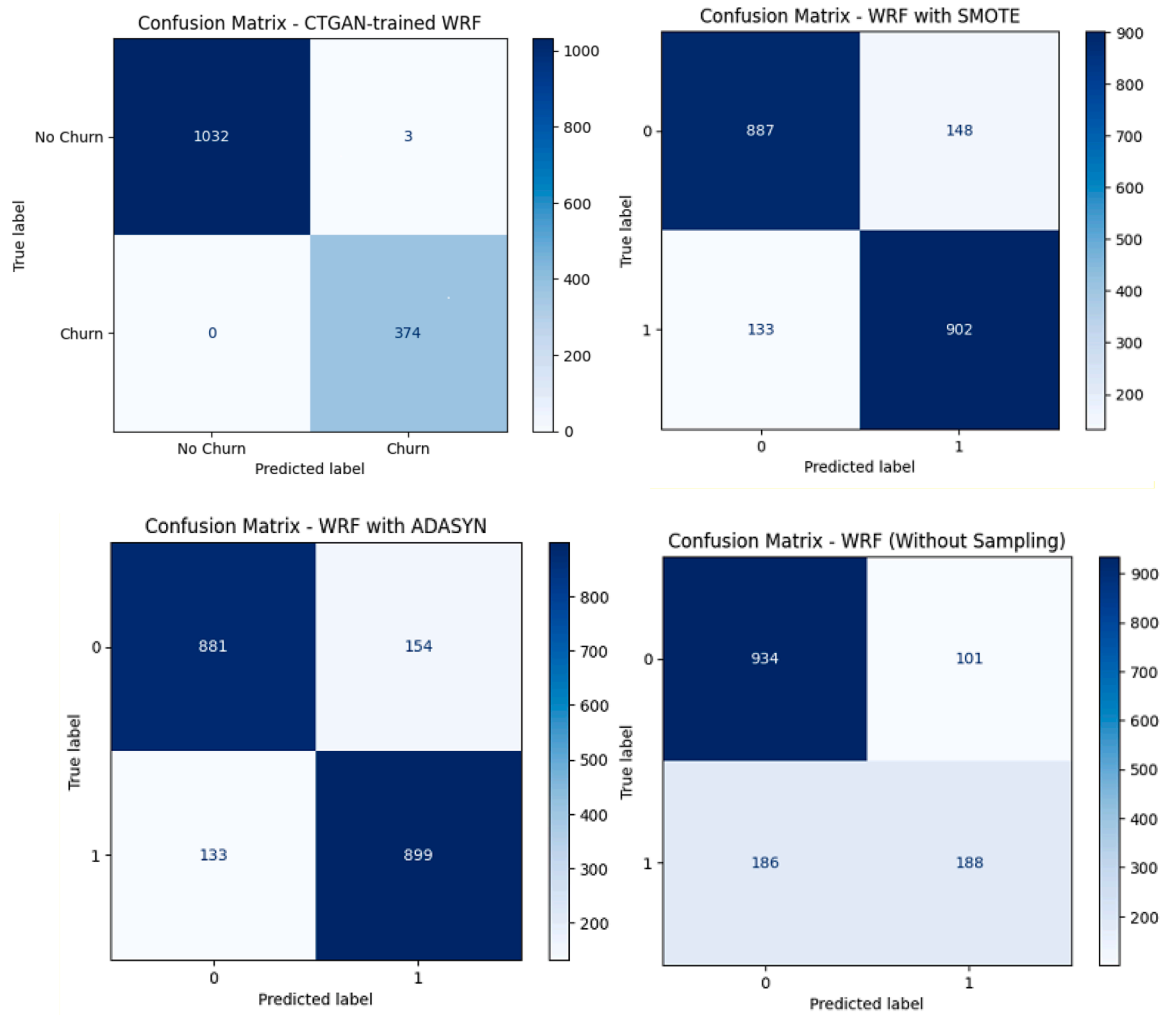


Fig. 4. Confusion matrices with various sampling methods.

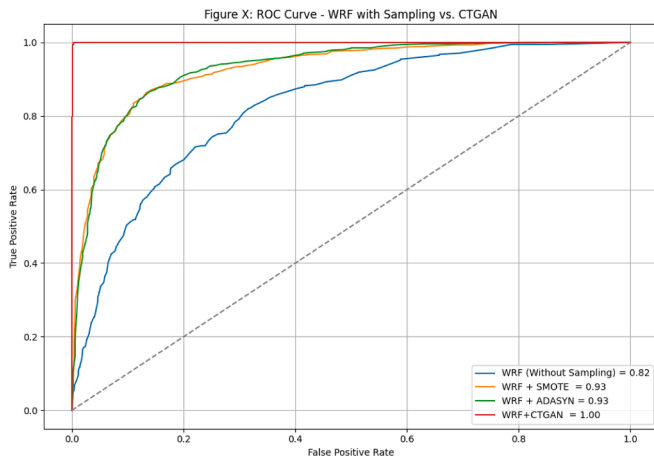


Fig. 5. ROC curve analysis of different sampling techniques.

citizens). The average customer tenure is about 32.37 months, ranging from 0 to 72 months, suggesting a mix of new and long-term users. Services like PhoneService have high availability (mean = 0.90), whereas optional features such as MultipleLines, InternetService, OnlineSecurity, OnlineBackup, and various streaming services exhibit more variability in usage, as reflected in their means and standard deviations.

Among continuous variables, MonthlyCharges average around \$64.76 with a standard deviation of \$30.09, spanning from \$18.25 to \$118.75. The Churn variable has a mean of 0.2654, indicating that approximately 26.5 % of customers in the dataset have churned. These statistics offer valuable insights into the distribution and central tendencies of the features, which are crucial for subsequent analysis and predictive modeling.

The bar chart in Fig. 2 illustrates the distribution of the target variable, Churn. It reveals a significant class imbalance in the dataset, with only 26.58 % of customers labeled as churners and 73.42 % as non-churners. This indicates that the majority of customers have remained with the service, underscoring the need for specialized techniques to address class imbalance during model training for accurate churn prediction.

To explore inter-variable relationships, a correlation matrix was constructed and visualized in Fig. 3. This graphical representation depicts the strength and direction of associations between call activity metrics and other customer attributes in the telecom dataset, offering insights into potential feature dependencies and multicollinearity.

8.2. Performance evaluation using the confusion matrix

The confusion matrices presented in Fig. 4 compare the performance of the Weighted WRF classifier under various data balancing techniques: CTGAN, SMOTE, ADASYN, and no sampling. The WRF model trained on CTGAN-synthesized data demonstrates the highest precision and recall,

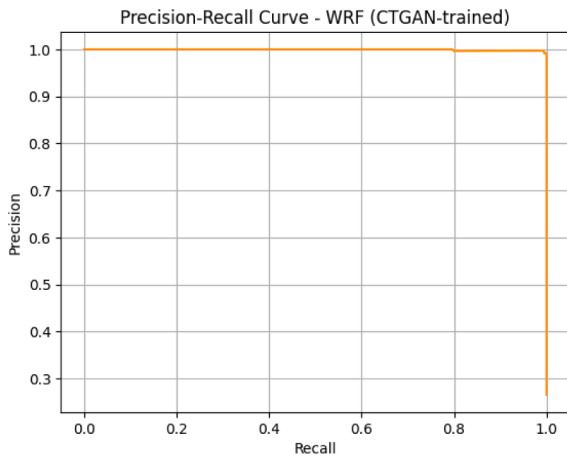


Fig. 6. PR Curve of CTGAN-trained WRF.

achieving near-perfect classification with only 3 false positives and no false negatives. In contrast, the WRF model trained without any sampling performs poorly, particularly in predicting churners, with 186 false negatives and only 188 true positives. Applying SMOTE leads to notable performance improvements, yielding 902 true positives and 887 true negatives, although 133 churners and 148 non-churners are still misclassified. Similarly, ADASYN produces results comparable to SMOTE, with 899 true positives and 881 true negatives, but slightly increases the number of false positives. Overall, the confusion matrix analysis highlights that CTGAN significantly outperforms other sampling methods in generating synthetic data for churn prediction, delivering superior model accuracy and reliability.

8.3. Analyzing sampling techniques using ROC curves

The ROC curves shown in Fig. 5 offer a visual comparison of the effectiveness of different sampling techniques applied to the WRF model for telecom churn prediction. Among the evaluated methods, the WRF model trained with CTGAN-generated synthetic data demonstrates the highest performance, achieving an AUC of 1.0000 indicating near-perfect classification between churners and non-churners. In contrast, the model trained without any sampling performs the poorest, with an AUC of just 0.8249, reflecting limited capability in handling the imbalanced dataset. Both SMOTE and ADASYN enhance the model's

discriminatory power, achieving AUCs of 0.9286 and 0.9310, respectively, with ADASYN showing a slight edge. These results clearly demonstrate that synthetic oversampling particularly with CTGAN significantly enhances classification performance in scenarios with severe class imbalance.

8.4. Precision-recall curve analysis

The Precision-Recall (PR) curve, on the other hand, is more informative when dealing with imbalanced datasets. It plots Precision (positive predictive value) against Recall (sensitivity). A high and relatively flat PR curve, as observed in the results (see Fig. 6), indicates that the model maintains high precision across a range of recall values. This suggests that the classifier is not only accurate but also effective in identifying the minority class likely the churned customers which is often the primary focus in churn prediction problems.

The classification report, based on the model trained using CTGAN-generated synthetic data in combination with the WRF, shows exceptional performance across both classes. Class 0 (non-churn) achieves perfect scores, with precision, recall, and F1-score all at 1.00, while Class 1 (churn) records nearly perfect values (precision = 0.99, recall = 1.00, F1-score = 1.00). This indicates that the model accurately identifies all churners with minimal false positives. The overall accuracy stands at 100 %, with macro and weighted averages further confirming the model's robustness despite the class imbalance. However, caution is warranted, as the use of synthetic data from CTGAN may lead to overfitting and artificially inflated performance metrics.

Classification Report of CTGAN-trained WRF precision recall f1-score support

```
0 1.00 1.00 1.00 1035
1 0.99 1.00 1.00 374 accuracy 1.00 1409 macro avg 1.00 1.00 1.00
1409 weighted avg 1.00 1.00 1.00 1409
```

8.5. Local interpretable model-agnostic explanations

The image is a LIME visualization explaining the model's prediction of "Not Churn" for a particular customer (see Fig. 7). Each bar represents the contribution of a specific feature to the prediction. Red bars indicate features that pushed the prediction toward churn, while green bars supported the prediction of not churn. For instance, features like Contract > 0.37, tenure > 0.92, and TotalCharges = 100.2 contributed negatively (red), suggesting churn risk. However, they were outweighed by features such as MonthlyCharges > 0.84, InternetService <= 1.00, and SeniorCitizen <= -0.44, which supported the prediction of not churn.

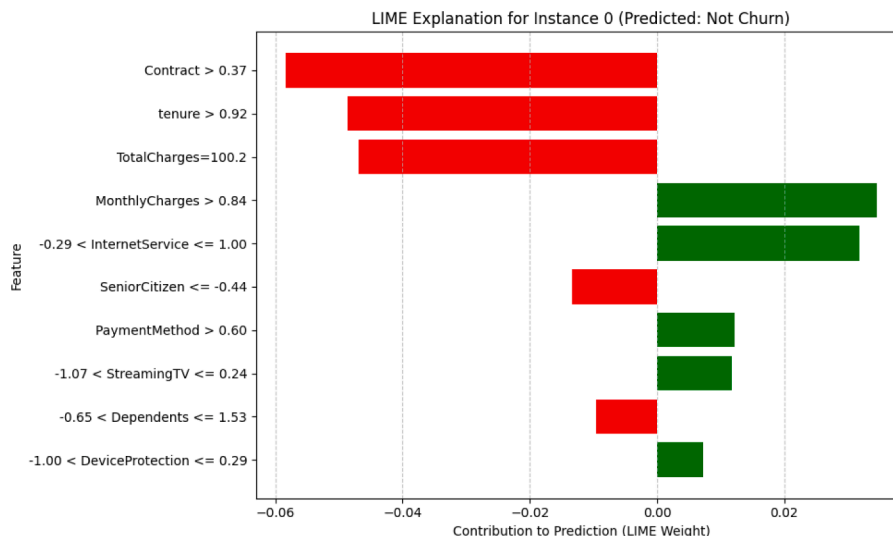


Fig. 7. LIME of CTGAN trained WRF.

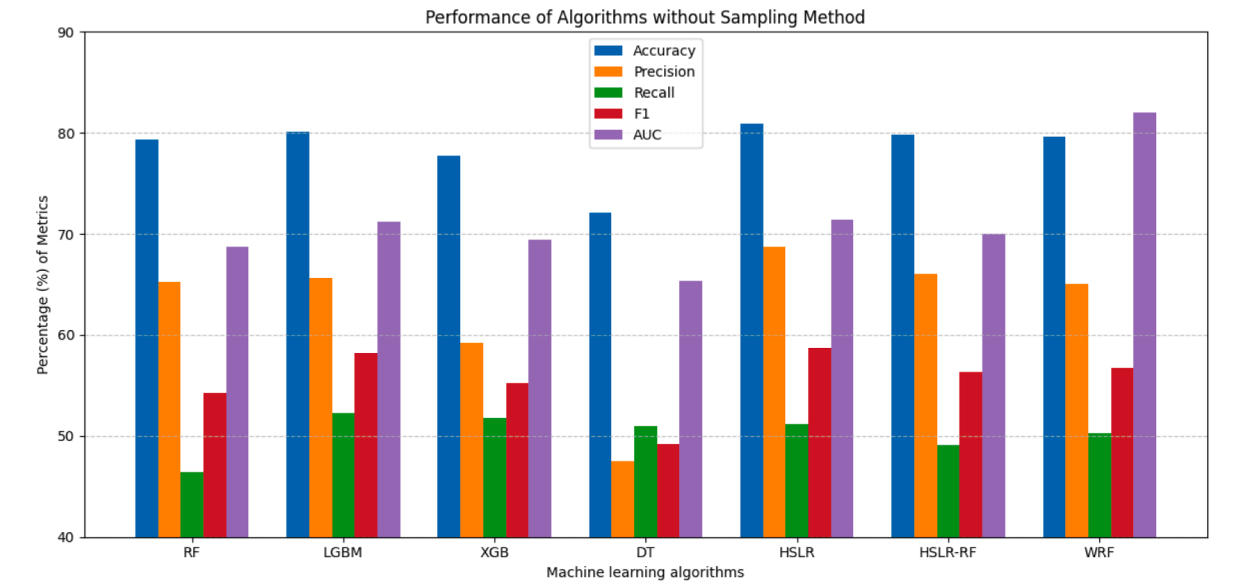


Fig. 8. Performance of algorithms without sampling.

Table 3
Algorithms performance evaluation with SMOTE.

Classifier	Accuracy	Precision	Recall	F1	AUC
RF	86.43 %	85.90 %	87.15 %	86.52 %	92.86 %
LGBM	84.98 %	84.67 %	85.41 %	85.04 %	93.62 %
XGB	84.64 %	84.44 %	84.93 %	84.93 %	93.10 %
DT	78.99 %	78.57 %	79.71 %	79.14 %	78.99 %
HSLR	85.80 %	84.92 %	87.05 %	85.97 %	93.46 %
HSLR-RF	86.43 %	85.90 %	87.15 %	86.52 %	92.86 %
WRF	86.43 %	85.90 %	87.15 %	86.52 %	92.86 %

stable range, and the use of DeviceProtection, which had a positive influence (green) supporting retention. Understanding these individual feature impacts empowers telecom providers to take targeted, proactive measures—such as offering loyalty incentives or enhanced support demonstrating how interpretability bridges AI predictions with actionable business strategies.

8.6. Performance of algorithms without sampling techniques

The evaluation of algorithm performance without sampling techniques highlights the superiority of hybrid and advanced ensemble methods. As detailed in Table 2, the HSLR model emerges as the best-performing classifier, achieving the highest accuracy (80.91 %) and

Table 4
Performance analysis of the algorithms using ADASYN.

Classifier	Accuracy	Precision	Recall	F1	AUC
RF	85.82 %	85.49 %	86.24 %	85.87 %	92.85 %
LGBM	84.29 %	84.67 %	85.14 %	85.40 %	93.53 %
XGB	84.52 %	84.10 %	85.08 %	84.59 %	93.03 %
DT	78.66 %	78.39 %	79.07 %	78.73 %	78.68 %
HSLR	82.24 %	80.42 %	85.17 %	82.73 %	88.24 %
HSLR-RF	82.49 %	79.65 %	87.21 %	83.26 %	90.25 %
WRF	86.31 %	85.84 %	86.92 %	86.37 %	92.90 %

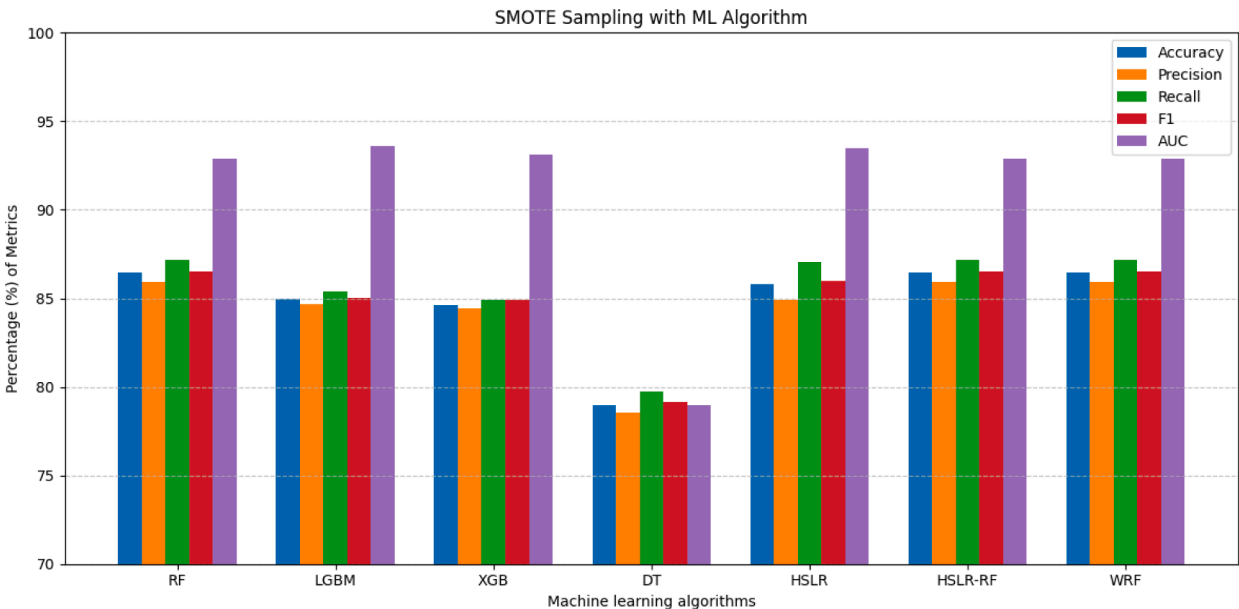


Fig. 9. Comparison of different algorithms using SMOTE.

Table 5
Performance metrics of the algorithms with CTGAN.

Classifier	Accuracy	Precision	Recall	F1	AUC
RF	99.65 %	99.73 %	98.93 %	99.33 %	100
LGBM	87.08 %	81.37 %	66.58 %	73.24 %	94 %
XGB	92.69 %	89.05 %	82.62 %	85.71 %	98 %
DT	99.57 %	99.73 %	98.66 %	99.19 %	100 %
HSLR	98.79 %	99.72 %	95.72 %	97.68 %	99 %
HSLR-RF	91.55 %	91.00 %	75.67 %	82.63 %	97 %
WRF	99.79 %	99.20 %	100.00 %	99.60 %	100 %

precision (68.71 %), reflecting its strong ability to predict positive outcomes with confidence and accuracy. With an F1 score of 58.68 % and an AUC of 71.40 %, it demonstrates balanced performance across metrics. LGBM follows closely, achieving 80.13 % accuracy, along with competitive precision (65.66 %), recall (52.28 %), and an F1 score of 58.21 %, suggesting it is a robust alternative. The WRF classifier, while delivering an impressive AUC of 82.00 %, shows slightly lower accuracy (79.63 %) and other metrics compared to HSLR. Fig. 8 visually summarizes these results, highlighting the superior performance of hybrid

and stacking methods such as HSLR and HSLR-RF, and their effectiveness in managing imbalanced datasets through advanced algorithmic design. This analysis underscores the utility of these sophisticated techniques, even in the absence of sampling strategies.

8.7. SMOTE enhanced classifier performance metrics

The effectiveness of ML algorithms with SMOTE sampling is compared in Table 3 and Fig. 9. Ensemble models such as RF, HSLR-RF, and WRF demonstrated superior performance across all metrics, each achieving 86.43 % accuracy, 85.90 % precision, 87.15 % recall, 86.52 % F1-score, and 92.86 % AUC. LGBM (AUC = 93.62 %) slightly outperformed XGB (AUC = 93.10 %) among the gradient boosting models, though both exhibited lower accuracy and F1-scores compared to the RF-based classifiers. DT showed the weakest results, with the lowest AUC (78.99 %), indicating poor handling of class imbalance even after SMOTE. The hybrid HSLR model also performed well, with an F1-score of 85.97 % and recall of 87.05 %, emphasizing its strength in detecting the minority class. The comparable outcomes of RF, WRF, and HSLR-RF highlight SMOTE's effectiveness in balancing class distribution,

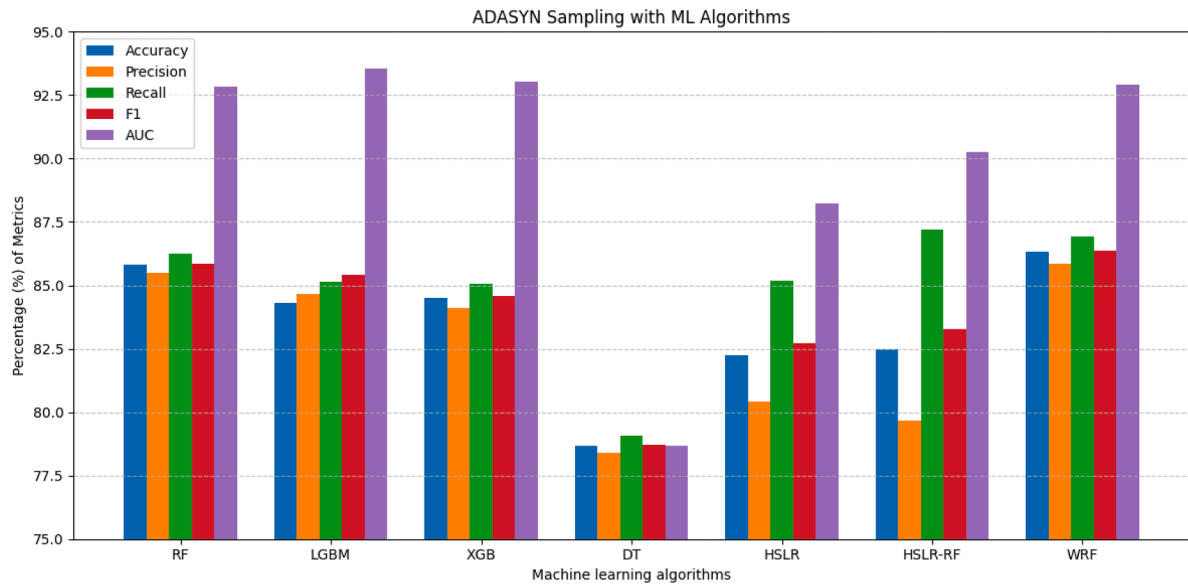


Fig. 10. Performance analysis: algorithms with ADASYN sampling.

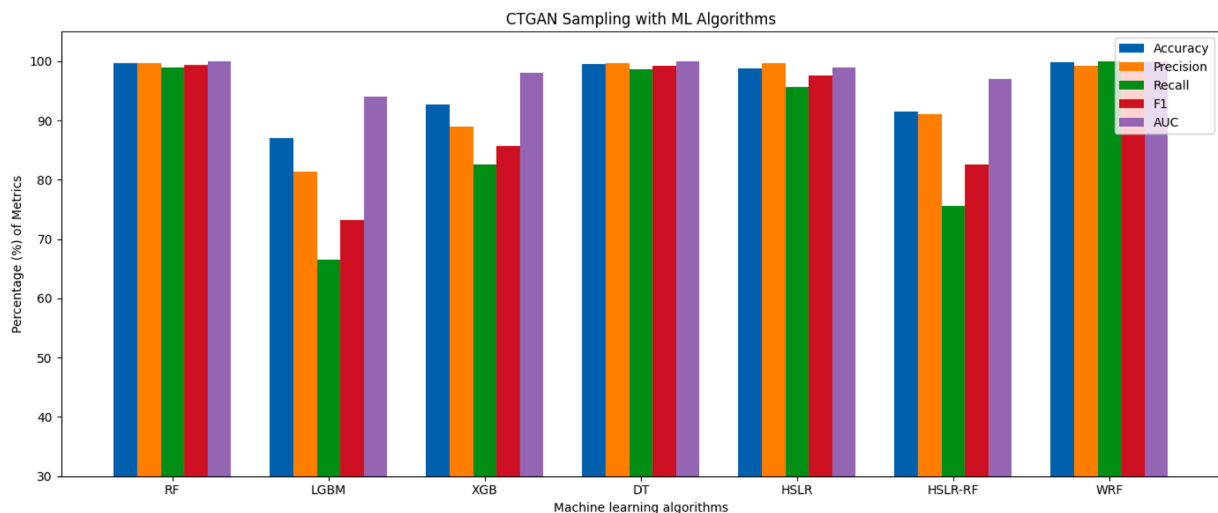


Fig. 11. Evaluating algorithms using CTGAN-enhanced methods.

Table 6
Comparative analysis of sampling techniques.

Classifier	Sampling Method	Accuracy	Precision	Recall	F1	AUC
RF	CTGAN	99.65 %	99.73 %	98.93 %	99.33 %	100 %
	SMOTE	86.43 %	85.90 %	87.15 %	86.52 %	92.86 %
	ADASYN	85.82 %	85.49 %	86.24 %	85.87 %	92.85 %
	Original	79.28 %	65.28 %	46.38 %	54.23 %	68.75 %
LGBM	CTGAN	87.08 %	81.37 %	66.58 %	73.24 %	94 %
	SMOTE	84.98 %	84.67 %	85.41 %	85.04 %	93.62 %
	ADASYN	84.29 %	84.67 %	85.14 %	85.40 %	93.53 %
	Original	80.13 %	65.66 %	52.28 %	58.21 %	71.22 %
XGB	CTGAN	92.69 %	89.05 %	82.62 %	85.71 %	98 %
	SMOTE	84.64 %	84.44 %	84.93 %	84.93 %	93.10 %
	ADASYN	84.52 %	84.10 %	85.08 %	84.59 %	93.03 %
	Original	77.79 %	59.20 %	51.74 %	55.22 %	69.45 %
DT	CTGAN	99.57 %	99.73 %	98.66 %	99.19 %	100 %
	SMOTE	78.99 %	78.57 %	79.71 %	79.14 %	78.99 %
	ADASYN	78.66 %	78.39 %	79.07 %	78.73 %	78.68 %
	Original	72.11 %	47.50 %	50.94 %	49.16 %	65.33 %
HSLR	CTGAN	98.79 %	99.72 %	95.72 %	97.68 %	99 %
	SMOTE	85.80 %	84.92 %	87.05 %	85.97 %	93.46 %
	ADASYN	82.24 %	80.42 %	85.17 %	82.73 %	88.24 %
	Original	80.91 %	68.71 %	51.21 %	58.68 %	71.40 %
HSLR-RF	CTGAN	91.55 %	91.00 %	75.67 %	82.63 %	97 %
	SMOTE	86.38 %	85.15 %	87.46 %	85.89 %	92.81 %
	ADASYN	82.49 %	79.65 %	87.21 %	83.26 %	90.25 %
	Original	79.84 %	66.06 %	49.06 %	56.31 %	69.99 %
WRF	CTGAN	99.79 %	99.20 %	100.00 %	99.60 %	100 %
	SMOTE	85.23 %	84.83 %	87.15 %	85.56 %	91.57 %
	ADASYN	86.31 %	85.84 %	86.92 %	86.37 %	92.90 %
	Original	79.63 %	65.05 %	50.27 %	56.71 %	82.00 %

reducing the added benefit of weighting in WRF. Overall, Fig. 9 reinforces the advantage of combining ensemble or hybrid models with SMOTE to optimize both sensitivity and specificity.

8.8. Impact of ADASYN on classifier performance analysis

The performance of different machine learning classifiers using ADASYN is summarized in Table 4 and visually represented in Fig. 9. Among them, the WRF achieved the best overall performance, with an

accuracy of 86.31 %, precision of 85.84 %, recall of 86.92 %, F1-score of 86.37 %, and an AUC of 92.90 %, demonstrating its strong capability in managing imbalanced data. Other ensemble methods, including RF and HSLR-RF, also performed competitively, closely aligning with WRF's results. Gradient boosting models such as LGBM and XGB showed the highest AUC values at 93.53 % and 93.03 %, respectively, indicating excellent class separation ability, though their accuracy and F1-scores slightly trailed behind the RF-based models. In contrast, the DT model recorded the lowest scores across all metrics, particularly with an accuracy of 78.66 % and an AUC of 78.68 %, revealing its limited effectiveness in addressing imbalance even after ADASYN resampling. Meanwhile, hybrid models like HSLR and HSLR-RF exhibited balanced recall and F1-scores, reflecting their sensitivity to detecting the minority class. Overall, the findings confirm that ensemble and hybrid classifiers especially WRF offer robust solutions for churn prediction when combined with the ADASYN technique.

8.9. Performance evaluation of classifiers enhanced by the CTGAN

Table 5 highlights significant variations in classifier performance, with WRF achieving the highest accuracy at 99.79 %, followed by RF (99.65 %) and DT (99.57 %), demonstrating the strength of ensemble methods. HSLR also performed well (98.79 %), while XGB (92.69 %) and HSLR-RF (91.55 %) delivered competitive results. LGBM trailed behind with 87.08 % accuracy, likely due to its sensitivity to noise and its leaf-wise growth strategy, which makes it prone to overfitting on CTGAN-generated synthetic data. Although CTGAN is effective in capturing complex relationships, it may introduce unrealistic samples, especially in the minority class, which LGBM interprets too aggressively. This leads to poor recall for the minority class, limiting its effectiveness in identifying actual churners. In contrast, models like RF and WRF handle such synthetic variations more robustly. Precision scores were highest for RF and DT (99.73 %), followed closely by HSLR (99.72 %) and WRF (99.20 %). WRF excelled in recall with a perfect 100 %, while RF and DT followed closely. F1 scores aligned with these results, with WRF at 99.60 %, RF at 99.33 %, and DT at 99.19 %. AUC scores (Fig. 10) confirmed perfect performance for RF, DT, and WRF (100 %), with strong showings from XGB (98 %), HSLR (99 %), and HSLR-RF (97 %). Overall, ensemble models—particularly WRF and RF—consistently outperformed others across all evaluation metrics Fig. 11.

8.10. Comparative analysis of sampling techniques

Table 6 presents a comparative analysis of various classifiers under different sampling techniques Original, SMOTE, ADASYN, and CTGAN evaluated using Accuracy, Precision, Recall, F1-score, and AUC. Overall, CTGAN-based synthetic data significantly enhanced classifier performance, with Weighted Random Forest (WRF) achieving the highest scores (Accuracy: 99.79 %, Recall: 100 %, AUC: 100 %), followed closely by Decision Tree (DT), Random Forest (RF), and Hybrid Statistical Linear Regression (HSLR), all showing strong results under CTGAN. Traditional resampling methods like SMOTE and ADASYN yielded noticeable improvements over the original imbalanced dataset particularly in Recall and F1-score but generally underperformed compared to CTGAN. Fig. 12 illustrates the accuracy achieved by each sampling technique, further emphasizing the superiority of CTGAN-enhanced models.

Interestingly, LGBM showed relatively lower performance with CTGAN, likely due to its sensitivity to noise in synthetic data. In contrast, ensemble and hybrid models such as WRF, RF, and HSLR consistently delivered robust and balanced results across all metrics. Both DT and RF achieved near-perfect AUC and high F1-scores when combined with CTGAN, indicating that tree-based models benefit significantly from high-quality synthetic data. Likewise, HSLR and HSLR-RF models demonstrated substantial gains in Recall and F1-score under CTGAN, confirming the model's strength in minority class detection even in

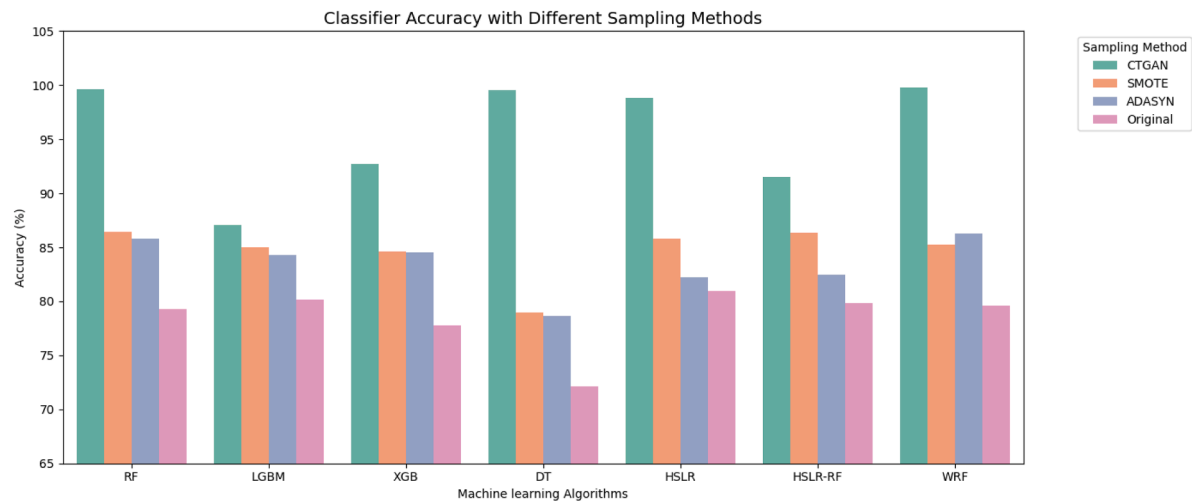


Fig. 12. Comparison of various sampling methods with accuracy.

Table 7
WRF performance with/without CTGAN (Mean ± SD, 95 % CI).

Metrics	CTGAN with WRF		WRF (Without Sampling)	
Accuracy	0.998 ± 0.001	95 % CI: [0.995, 1.000]	0.797 ± 0.010	95 % CI: [0.777, 0.816]
Precision	0.992 ± 0.005	95 % CI: [0.982, 1.000]	0.651 ± 0.028	95 % CI: [0.597, 0.702]
Recall	1.000 ± 0.000	95 % CI: [1.000, 1.000]	0.503 ± 0.025	95 % CI: [0.455, 0.551]
F1 Score	0.996 ± 0.002	95 % CI: [0.991, 1.000]	0.567 ± 0.022	95 % CI: [0.523, 0.609]
AUC	1.000 ± 0.000	95 % CI: [0.999, 1.000]	0.825 ± 0.012	95 % CI: [0.802, 0.846]

hybrid configurations. For instance, Recall for SMOTE + RF reached 87.15 %, while CTGAN + RF improved it to 98.93 %. Lastly, the original imbalanced dataset consistently resulted in the poorest performance across all classifiers, particularly affecting Recall and AUC, thus reinforcing the critical need to address class imbalance in churn prediction tasks.

8.11. Evaluating model performance with mean ± SD and 95 % confidence intervals

The performance comparison between CTGAN with WRF and WRF without any sampling technique is comprehensively summarized using key evaluation metrics accuracy, precision, recall, F1-score, and AUC each reported with its respective standard deviation and 95 % confidence interval (CI), thereby reflecting both central tendency and variability. Table 7 presents this comparison, revealing a clear advantage for the CTGAN-enhanced model. CTGAN+WRF achieves near-perfect performance: an accuracy of 0.998 ± 0.001 , precision of 0.992, recall of 1.000, F1-score of 0.996, and AUC of 1.000 all with tight confidence intervals, indicating consistent and reliable predictions. In contrast, WRF on the original imbalanced dataset shows substantially lower results: accuracy (0.797), precision (0.651), recall (0.503), F1-score (0.567), and AUC (0.825), reflecting notable misclassifications. These findings confirm the effectiveness of CTGAN-generated synthetic data in significantly enhancing model performance, particularly in imbalanced classification scenarios, as evidenced by superior mean scores, lower variability, and tighter confidence intervals across all metrics.

9. Conclusion

This paper presents a systematic comparison of sampling techniques

SMOTE, ADASYN, and CTGAN integrated with ML classifiers to address class imbalance in telecom churn prediction. The results show that the CTGAN combined with WRF significantly outperforms other configurations, achieving an accuracy of 99.79 %, perfect recall, and an AUC of 1.0 for the minority class. These outcomes highlight CTGAN’s effectiveness in generating high-quality synthetic data that enhances the model’s ability to identify churners. To support reproducibility, the dataset is thoroughly documented, including the class imbalance ratio, total sample size, and feature type distribution providing context for the challenges of imbalanced classification in real-world telecom settings. While the results are highly promising, their practical deployment necessitates validation in operational environments, accounting for CRM system integration, compliance with data privacy regulations, and real-time processing capabilities. To enhance interpretability and support informed decision-making, the study incorporates LIME, enabling telecom operators to understand key drivers behind individual predictions and design targeted, explainable retention strategies particularly useful for non-technical stakeholders.

Future directions

- Advanced Generative Models: Explore Variational Autoencoders (VAEs), Diffusion Models, and hybrid architectures to improve the realism and utility of synthetic data, including conditional generation for domain-specific constraints such as demographic profiles.
- Real-Time Systems: Develop streaming-compatible architectures capable of low-latency prediction, while systematically evaluating trade-offs between batch and online learning.
- Interpretability & Scalability: Extend interpretability tools such as LIME and SHAP to handle temporal, sequential, and multi-modal data, and validate their robustness across multiple telecom operators and large-scale deployments.
- Continuous Learning: Integrate active learning and adaptive sampling methods to manage concept drift and incorporate feedback from ongoing retention efforts for continuous model refinement.
- Ethical AI: Emphasize privacy-preserving techniques and fairness-aware sampling strategies to minimize bias and promote responsible AI use.

This study lays a strong foundation for next-generation churn prediction systems, emphasizing the critical importance of real-world validation, interpretability, scalability, and ethical AI to ensure meaningful and practical impact in the telecommunications sector.

CRediT authorship contribution statement

B. Shunmuga Priya: Writing – original draft. **G. Chitra:** Supervision. **R. Ramalakshmi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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